

Swaroop Naik

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Research interests: model robustness & distribution shift · LLM evaluation · retrieval-augmented generation · applied ML

EDUCATION

B.Tech, Computer Engineering — Vidyalankar Institute of Technology, Mumbai Aug 2023 – May 2027 (expected)
GPA: 9.42/10 · **Relevant coursework:** Machine Learning, Deep Learning, Computer Vision, NLP, Retrieval-Augmented Generation, Data Structures & Algorithms, Database Management, OOP, Computer Architecture, Calculus

RESEARCH PROJECTS

Cross-Site Generalization of ML Models for Solar PV Power Prediction Aug 2026
Open benchmark · Python, CatBoost, XGBoost, scikit-learn · code

- Built a reproducible cross-continent benchmark of how PV power models transfer across panel technology, climate and a four-year time gap, using public DKASC (Australia) and NIST (USA) data; normalized output to capacity factor so 5 kW–217 kW arrays are directly comparable.
- Evaluated six regressors within-site and zero-shot cross-site over an all-pairs matrix with five random seeds.
- Showed cross-technology and temporal shift are statistically indistinguishable from lossless (mean $\Delta R^2 = 0.001$), while cross-climate shift is ≈18× larger (mean $\Delta R^2 = 0.024$, Mann–Whitney $p \approx 2\times10^{-10}$).
- Compared five domain-adaptation methods: unsupervised alignment (CORAL, importance weighting) did not help, while a fine-tuned neural encoder recovered ≈98% of the cross-climate gap with only 20% of target-site data.
- Extended to fault detection on GPVS-Faults: near-perfect within-mode classification (F1 = 0.999) degrades across MPPT/IPPT operating modes.

Agentic LLM Evaluation System 2026
CrewAI, Ollama, LLaMA 3, Mistral, phi3, qwen3.5:9b, Sentence-Transformers, SQLite · code

- Built a local multi-agent pipeline in which two models answer, a verifier fact-checks, a judge scores both answers on an 8-criterion rubric and a combiner synthesizes the result, with a six-signal confidence score; completed a 5,000-question judged dataset.
- Found the initial judge (CompassJudger-1) gave identical scores across all 8 criteria despite healthy downstream metrics, and replaced it; showed fixed answer order skewed win rates to 60–72% for one position, falling to 43/57 once order was randomized.
- Removed a circular confidence signal ($r = 0.737$ with the judge score) and added a control group to validate the retry threshold: sub-threshold answers improved on retry 66.7% of the time ($n = 144$) vs. 54.7% for controls ($n = 53$).

EXPERIENCE

Machine Learning Consultant — Reliance Industries Ltd. May 2026 – Jul 2026 · Mumbai
MatAI — natural-language analytics over enterprise material-management reports

- Built an AI assistant letting managers and engineers query 100+ enterprise reports in plain English with no SQL or schema knowledge: FastAPI backend, Angular front end, local LLM inference, in-memory columnar query engine; deployed on an internal server with SSO into the existing portal.
- Replaced direct LLM-generated SQL with a typed query plan validated against the live schema before parameterized SQL is emitted, removing injection via user input and hallucinated column names; added filter accumulation across conversational turns.
- Cut latency from 60–90 s to <1 s for ~40% of queries via fast-path interception, schema caching and parallel retrieval; benchmarked candidate LLMs on real workloads, rejecting one that reported figures absent from the data.

Software Developer Intern (Gen AI) — Ksolves India Ltd. May 2025 – Jul 2025 · Pune
PNM AI-Powered Helpdesk Assistant

- Built a chat-based assistant for network-maintenance queries with a RAG pipeline over FAISS/Chroma vector stores (LangChain, Streamlit), supporting PDF and image inputs.
- Added thread summarization and metadata summaries for multi-turn context management, and PostgreSQL session logging of queries, responses and summaries.
- Built LangChain applications and internal experiments involving prompt engineering, output parsing, multi-step pipelines and agent design.

OTHER PROJECTS

E-commerce Analytics at Scale 2026

- Apache Spark pipeline over 1,067,371 real transactions feeding a SQLite data mart, Express API and React dashboard; RFM segmentation, K-Means and FP-Growth; churn classifier AUC 0.803, spend regressor R^2 0.41 (log).

Prostate Cancer Risk Predictor 2026

- Hybrid neural network (Keras) and fuzzy-logic system (scikit-fuzzy) over six biomarker inputs, so predictions come with clinician-readable rules; served via FastAPI.

University Timetable Scheduler Sep 2025

- Full-stack scheduling system (Next.js 14, TypeScript, MongoDB, NextAuth.js) with role-based access; Mongoose transactions make slot selection atomic, preventing double-booking under concurrent requests.

Farmer's Basket — Smart India Hackathon 2024 Aug 2024 – Nov 2024

- Marketplace for farmers to sell produce directly and rent or buy equipment, with a plain-language guide to government schemes (problem statements Q-1637 and Q-1690); cleared the college round.

LEADERSHIP

Vice Chairperson — Computer Engineering Students' Association (CESA), VIT 2025 – 2026

- Coordinated participants, teams and volunteers for HACKFEST: INNOV8 TMRW, a national hackathon with 180+ team registrations from several states, alongside logistics and on-ground execution.
- Ran the association's technical programme, including Cognition Clash (AI/ML competition) and the Agentic AI Workshops, keeping coordinators, volunteers and faculty to a shared schedule; recognized by a departmental Letter of Appreciation.

SKILLS

Languages	Python, Java, C, SQL, R, JavaScript/TypeScript
ML & Data	PyTorch, Keras, scikit-learn, XGBoost, CatBoost, Pandas, NumPy, Apache Spark (MLlib)
LLM Systems	LangChain, CrewAI, Ollama, FAISS, Chroma, Sentence-Transformers, RAG, LLM-as-judge evaluation
Engineering	FastAPI, Flask, Django, Node.js/Express, React, Next.js, Streamlit, PostgreSQL, MongoDB, SQLite, Docker

AWARDS & CERTIFICATIONS

- First Runner-Up, Livelihoods India Case Study Competition 2024 — case study on reCharkha, a social enterprise upcycling plastic waste and employing rural women.
- Certificate of Merit in Python Programming (Semester 3).
- NVIDIA: Getting Started with Deep Learning · Coursera: Supervised Machine Learning: Regression and Classification; Crash Course on Python · NPTEL: Programming in Java · Postman API Fundamentals Student Expert · Interactive Dashboards with Streamlit and Python.